The Impact of Paid Generative AI on Quality of Life: A Poverty Premium Perspective

Wei-Lun Chang, Yan-Yi Huang
Department of Business Administration
National Taipei University
New Taipei City, Taiwan
wlchang@gm.ntpu.edu.tw; yanyi110666@gmail.com

Abstract*—Generative AI subscription services are reshaping how users access digital resources and productivity tools. However, limited research has examined how paid users transform their actual use experience into improvements in quality of life. Integrating UTAUT2 and the Expectation-Confirmation Model, this study investigates paid generative AI service users in the United Kingdom. Based on 819 valid survey responses, the proposed model was analyzed using partial least squares structural equation modeling. The results show that performance expectancy, price value, and utilitarian motivation significantly promote use behavior. Use behavior further strengthens confirmation, which subsequently improves users' perceived quality of life. From the poverty premium perspective, this study argues that digital inequality in AI subscription services is not only about whether lower-income users can afford subscription fees, but also about whether paid services return sufficient value after adoption. Although income did not emerge as a dominant explanatory factor in the final model, this finding suggests that the poverty premium in paid GenAI services should not be understood only as an affordability gap. Instead, it may also operate as a usage-to-outcome gap, where users pay for the same service but differ in their ability to convert paid access into meaningful quality-of-life benefits.

***Index Terms*—Generative AI, subscription services, quality of life, poverty premium, UTAUT2, ECM.**

I. INTRODUCTION

Generative Artificial Intelligence (GenAI) services, such as ChatGPT, Copilot, Gemini, and Notion AI, have rapidly expanded across education, workplaces, and daily life. Many operate under a freemium model, offering basic access for free while requiring subscription payments for advanced functions. In the UK, higher-income and digitally capable users are more likely to benefit from digital tools [1], while younger users report higher GenAI adoption than older groups [2], [3]. This raises concerns that paid GenAI services may reinforce unequal access.

This issue is closely related to the poverty premium. UK studies show that low-income households often pay more for essentials such as energy, telecommunications, insurance, and credit [4]-[7]. Recent evidence also shows higher utility costs among poorer households [8]. As digital services increasingly rely on subscriptions and tiered access, similar inequalities may appear in the AI market, as low-income users may pay a higher relative cost for GenAI access or be excluded from productivity-enhancing tools [9], [10].

Existing research remains incomplete. AI adoption studies emphasize trust, perceived usefulness, and algorithm aversion [11], [12], while freemium research focuses on pricing and conversion in non-AI contexts [13]-[16]. Digital inequality studies show that income, education, and age shape digital benefits [17]-[20], but rarely examine GenAI subscriptions as a possible digital poverty premium. This study integrates UTAUT2 [21] and ECM [22] to examine how paid GenAI users' perceptions shape actual use behavior, how use behavior enhances confirmation, and whether confirmation improves perceived quality of life. From a poverty premium perspective, this study further examines whether income changes the relationship between price value and use behavior.

II. THEORETICAL BACKGROUND AND HYPOTHESES

UTAUT2 explains how users form behavioral intentions toward technology based on perceived usefulness, ease of use, available support, and value for money [21]. This study adopts performance expectancy, effort expectancy, facilitating conditions, price value, and utilitarian motivation as antecedents of use behavior for premium AI services. ECM explains post-adoption evaluation [22], where actual use influences confirmation and confirmation subsequently affects quality of life.

Performance expectancy reflects the degree to which users believe a technology enhances task performance. Users who perceive stronger productivity, learning, or work-related benefits are more likely to actively use the service [24]-[26]. Effort expectancy reflects perceived ease of learning and use; lower effort can reduce cognitive barriers to invest in a service [27], [28]. Facilitating conditions refer to the resources, knowledge, infrastructure, and support needed for effective use; these conditions may affect both willingness to pay and actual use behavior [29], [30]. Price value captures whether perceived benefits outweigh monetary costs [31], [32]. Utilitarian motivation refers to goal-oriented and productivity-driven use [33], [34]. Income is used to test whether paid GenAI subscriptions create a value-realization gap, where users pay for the same service but differ in their ability to convert paid access into quality-of-life benefits [4]–[8], [17]–[20]. AI literacy refers to users' ability to understand, evaluate, and effectively interact with AI technologies. Users need sufficient AI-related knowledge and skills to select appropriate functions, formulate prompts, evaluate outputs, and apply AI-generated results to real tasks [35]–[37]. Based on these arguments, this study proposes hypotheses linking UTAUT2 constructs to use behavior, ECM constructs to confirmation and quality of life, and income and AI literacy as moderating variables.

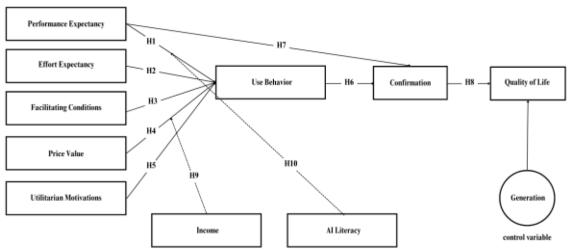

Fig. 1. Research model

III. METHODOLOGY

A. Sample and Data Collection

This study adopted a questionnaire survey method and targeted UK users who had previously used free AI services and later subscribed to paid versions. A screening question ensured that only respondents meeting both criteria proceeded to the survey. Data were collected through a third-party online research company. A total of 1,043 responses were collected, and 220 invalid responses were removed due to excessive identical answers, resulting in 819 valid samples.

B. Measurement and Analysis

The questionnaire was developed from validated scales on UTAUT2, ECM, use behavior, AI literacy, Income and quality of life. Generation was included as a control variable predicting perceived quality of life and was dummy-coded, with Gen X coded as 0 and Xennials coded as 1. All items were measured using a seven-point Likert scale. Quality of life was adapted from WHOQOL-BREF [23] and captured psychological well-being and social relationships. The data were analyzed using PLS-SEM with SmartPLS 3. The measurement model assessed reliability, convergent validity, and discriminant validity. The structural model examined path coefficients, significance levels, R^2 values, and model quality.

TABLE I. SAMPLE PROFILE

Category	Gen X (n=401)	Xennials (n=418)
Male	207 (51.6%)	184 (44.0%)
Female	194 (48.4%)	234 (56.0%)
Secondary or below	223 (55.6%)	155 (37.1%)
Bachelor's	117 (29.2%)	150 (35.9%)
Master's or doctorate	61 (15.2%)	113 (27.0%)
Full-time employment	264 (65.8%)	305 (73.0%)
Premium use < 3 months	173 (43.1%)	187 (44.7%)

IV. RESULTS

A. Measurement Model

During measurement assessment, UM1 was removed due to insufficient factor loading, while FC3、FC4、QoL3、QoL4、QoL6、QoL7、QoL8 were excluded because of discriminant validity concerns. The remaining items showed satisfactory reliability and validity. Factor loadings exceeded 0.70, Cronbach's alpha and composite reliability were above 0.70, and AVE values exceeded 0.50 for all constructs. HTMT values were below 0.90, and the Fornell-Larcker criterion was satisfied, confirming acceptable discriminant validity. Model fit was acceptable, with SRMR = 0.077. Common method bias was assessed using Harman's single-factor test, which showed that the first factor accounted for 57.76% of the total variance. Although this exceeded the 50% threshold, supplementary full collinearity VIF values ranged from 2.827 to 3.785 and remained below 5.0, suggesting that common method bias was not severe but should be acknowledged as a limitation.

B. Structural Model

Collinearity was assessed using inner VIF values. The VIF values ranged from 1.002 to 3.508, all below the conservative threshold of 5.0, indicating that multicollinearity was not a serious concern. The model demonstrated substantial explanatory power, with R^2 values of 0.689 for use behavior, 0.629 for confirmation, and 0.549 for quality of life. The estimated model also showed an acceptable model fit, with an SRMR value of 0.076, which was below the recommended threshold of 0.08.

AI literacy had the strongest effect on use behavior (β = 0.308, $p < 0.001$), followed by performance expectancy (β = 0.245, $p < 0.001$), utilitarian motivation ($\beta = 0.226$, $p < 0.001$), and price value ($\beta = 0.160$, $p < 0.001$). In contrast, effort expectancy and facilitating conditions did not reach the conventional 5% significance level. Income also did not significantly affect use behavior.

For the post-adoption process, use behavior significantly enhanced confirmation ($\beta = 0.518$, $p < 0.001$), and confirmation strongly improved quality of life ($\beta = 0.733$, $p < 0.001$). Performance expectancy also had a significant direct effect on confirmation ($\beta = 0.332$, $p < 0.001$). These findings indicate that paid GenAI subscriptions improve users' perceived quality of life primarily through a post-adoption mechanism in which actual use strengthens confirmation, and confirmation subsequently translates into quality-of-life improvement.

Regarding the additional effects, generation had a significant positive effect on quality of life ($\beta = 0.083$, $p < 0.001$). The moderating effect of AI literacy on the relationship between performance expectancy and use behavior was significant but negative ($\beta = -0.039$, $p < 0.05$), indicating that the effect of performance expectancy on use behavior was weaker among users with higher AI literacy. However, the moderating effect of income was not significant ($\beta = 0.018$, $p > 0.05$).

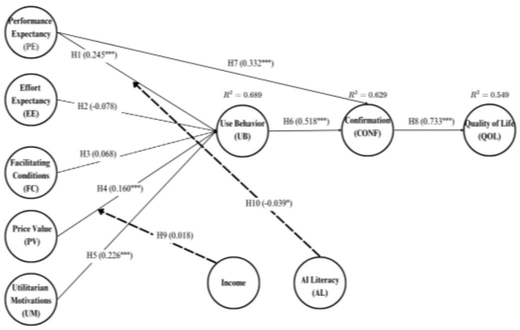

Fig 2. Structural Model Assessment Results

TABLE II. MAIN STRUCTURAL FINDINGS

Path	β	p-value	Result
PE -> UB	0.245	< .001	significant
EE -> UB	-0.078	0.055	Not significant
FC -> UB	0.068	0.062	Not Significant
PV -> UB	0.160	< .001	Significant
UM -> UB	0.226	< .001	Significant
UB -> CONF	0.518	< .001	Significant
PE -> CONF	0.332	< .001	Significant
CONF -> QoL	0.733	< .001	Significant
Generation → QoL	0.083	< .001	Significant
Income moderation	0.018	> 0.05	Not Significant
AI literacy moderation	-0.039	> 0.05	significant, negative

V. DISCUSSION

Paid GenAI subscriptions improve perceived quality of life through a post-adoption mechanism rather than through paid access alone. Use behavior significantly strengthened confirmation, and confirmation strongly improved quality of life. This finding suggests that users do not benefit from paid GenAI services simply because they subscribe to them. Instead, quality-of-life improvement depends on whether actual use confirms that the service meets users' expectations and provides meaningful value.

Second, actual use behavior among paid GenAI users is mainly driven by value- and capability-based factors. AI literacy, performance expectancy, utilitarian motivation, and price value significantly increased use behavior, while effort expectancy and facilitating conditions did not reach the conventional level of significance. This indicates that, for users who have already upgraded to paid GenAI services, continued use is shaped less by basic ease of use or external support and more by whether users perceive the service as useful, practical, worth its cost, and whether they have the literacy needed to use AI effectively.

Third, the findings refine the poverty premium perspective in the context of paid GenAI subscriptions. Income did not emerge as the dominant explanatory factor in the final model, suggesting that inequality in this context may not operate only as an affordability gap. Since all respondents had already moved from free to paid GenAI services, the more important issue is whether paid access can be converted into meaningful outcomes. Thus, this study extends the poverty premium discussion toward a usage-to-outcome gap: users may pay for the same AI service, but the extent to which they benefit depends on actual use, confirmation, and the ability to realize post-adoption value.

VI. CONCLUSION

This study examined how users who had upgraded from free to paid GenAI services. They converted their paid access into use experience and perceived quality-of-life outcomes. The findings suggest that the value of paid GenAI subscriptions is mainly formed after adoption. AI literacy, performance expectancy, utilitarian motivation, and price value. Associated with actual use behavior, use behavior then increased confirmation which further improved perceived quality of life. These results indicate that paid access itself is not sufficient. Users need to actively use the service and confirm its value before paid GenAI subscriptions can contribute to quality-of-life improvement.

This study extends UTAUT2 and the Expectation-Confirmation Model to the paid GenAI subscription context by shifting the focus from willingness to pay to post-adoption value realization. It also provides a different way to understand the poverty premium in AI subscription services. In this context, inequality may not only appear as an affordability gap before payment, but also as a usage-to-outcome gap after payment. Users may pay for the same service, but the extent to which they benefit depends on whether paid access can be converted into actual use, confirmation, and meaningful outcomes.

For GenAI service providers, the findings suggest that increasing subscriptions should not be the only goal. Providers also need to help users realize value after payment. Clear onboarding, task-based usage guidance, transparent pricing, and AI literacy support may help users better understand how to apply paid GenAI tools to work, learning, and daily tasks. These measures are especially important for users with fewer digital or socioeconomic resources, because they may face greater difficulty converting paid access into practical benefits.

REFERENCES

[1] Lloyds Banking Group, UK consumer digital index 2025, Lloyds Banking Group, 2025.

[2] KPMG UK, Generative AI in the workplace: Skills, learning and transformation in the UK, KPMG UK, 2024.

[3] Zuora and The Harris Poll, AI trends: How consumers are using and paying for generative AI services, 2024.

[4] Save the Children, The UK poverty rip-off: The poverty premium 2010, Save the Children, 2011.

[5] D. Hirsch, M. Padley, and L. Valadez, The poverty premium: When low-income households pay more, Loughborough University, 2016.

[6] M. Davies, A. Finney, Y. Hartfree, M. Padley, and S. Collard, Measuring the poverty premium, University of Bristol, 2016.

[7] S. Corfe, The poverty premium: When low-income households pay more, Social Market Foundation, 2018.

[8] F. Rasanga, T. Harrison, and R. Calabrese, "Measuring the energy poverty premium in Great Britain," Energy Economics, vol. 136, p. 107726, 2024.

[9] Digital Poverty Alliance, Digital poverty in the UK: Evidence review, 2022.

[10] Ofcom, Online nation 2022 report, Ofcom, 2022.

[11] B. J. Dietvorst, J. P. Simmons, and C. Massey, "Algorithm aversion," Journal of Experimental Psychology: General, vol. 144, no. 1, pp. 114-126, 2015.

[12] C. Longoni, A. Bonezzi, and C. K. Morewedge, “Resistance to medical artificial intelligence,” Journal of Consumer Research, vol. 46, no. 4, pp. 629-650, 2019.

[13] X. Gu, “When to charge for online content?” Journal of Media Economics, vol. 31, no. 2, pp. 63-81, 2018.

[14] A. Lambrecht and K. Misra, “Fee or free,” Management Science, vol. 63, no. 4, pp. 1150-1165, 2017.

[15] R. Iyengar, K. Jedidi, and H. Koh, “Paywalls and subscriptions,” Marketing Science, vol. 41, no. 1, pp. 71-95, 2022.

[16] H. Kato, “Pricing strategies for digital subscriptions,” Journal of Media Business Studies, vol. 19, no. 2, pp. 141-163, 2022.

[17] A. J. A. M. van Deursen, “Digital inequality during a pandemic,” Journal of Medical Internet Research, vol. 22, no. 8, e20073, 2020.

[18] E. Hargittai, “Potential and pitfalls of survey research on digital inequality,” Communication Methods and Measures, vol. 15, no. 2, pp. 92-99, 2021.

[19] M. Büchi, N. Festic, and M. Latzer, “How social well-being is affected by digital inequalities,” International Journal of Communication, vol. 12, pp. 3686-3706, 2018.

[20] M. H. Nguyen, E. Hargittai, and W. Marler, “Digital inequality in communication and information,” Information, Communication & Society, vol. 24, no. 2, pp. 236-254, 2021.

[21] V. Venkatesh, J. Y. L. Thong, and X. Xu, “Consumer acceptance and use of information technology,” MIS Quarterly, vol. 36, no. 1, pp. 157-178, 2012.

[22] A. Bhattacherjee, “Understanding information systems continuance,” MIS Quarterly, vol. 25, no. 3, pp. 351-370, 2001.

[23] World Health Organization, WHOQOL-BREF: Introduction, administration, scoring and generic version of the assessment, 1996.

[24] K. He, J. Zhang, and Y. Zeng, “Households’ willingness to pay for energy utilization of crop straw,” Energy Policy, vol. 140, p. 111373, 2020.

[25] L. Chen et al., “Determinants of consumer’s purchase intention on fresh e-commerce platform,” SAGE Open, vol. 11, no. 2, 2021.

[26] C.-C. Wu et al., “Understanding older adults’ purchase intentions toward smartwatches,” Journal of Applied Gerontology, 2025.

[27] W. Cao and L. Putit, “Influence of effort expectancy on consumers’ purchase intention,” International Journal of Business and Society, 2024.

[28] M. Zhang, H. Hassan, and M. W. Migin, “Online community group buying platform during pandemic,” Sustainability, vol. 15, no. 3, p. 2433, 2023.

[29] Yuniaristanto et al., “Exploring determinants of intention to purchase electric motorcycles,” Transportation Research Part F, vol. 100, pp. 475-492, 2024.

[30] M. Mashayekhy, F. Nosrati, and M. Ghasemaghaei, “Unlocking the future of education: Empirical insights into the adoption of generative AI in higher education,” Communications of the Association for Information Systems, vol. 57, no. 1, pp. 1270–1295, 2025.

[31] J. Song, S. Ryu, and Y. Kim, “Free to premium in mobile TV service,” Asia Pacific Journal of Information Systems, vol. 33, no. 2, pp. 318-341, 2023.

[32] Y. Wang et al., “Consumer perceived value and MOOC subscription intentions,” Journal of Computer Information Systems, 2025.

[33] S.-C. Chang, Y.-F. Lin, and Y.-P. Chiu, “From freemium to premium app services,” International Journal of Marketing, Communication and New Media, vol. 11, no. 21, pp. 112-131, 2023.

[34] Jo, “Uncovering the reasons behind willingness to pay for ChatGPT-4 Premium,” Journal of Retailing and Consumer Services, vol. 76, p. 103762, 2025.

[35] D. Long and B. Magerko, “What is AI literacy? Competencies and design considerations,” in Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, Honolulu, HI, USA, 2020, pp. 1–16.

[36] D. T. K. Ng, J. K. L. Leung, S. K. W. Chu, and M. S. Qiao, “Conceptualizing AI literacy: An exploratory review,” Computers and Education: Artificial Intelligence, vol. 2, Article 100041, 2021.

[37] S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, J. Teevan, R. Kikin-Gil, and E. Horvitz, “Guidelines for human-AI interaction,” in Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, Glasgow, Scotland, UK, 2019, pp. 1–13.